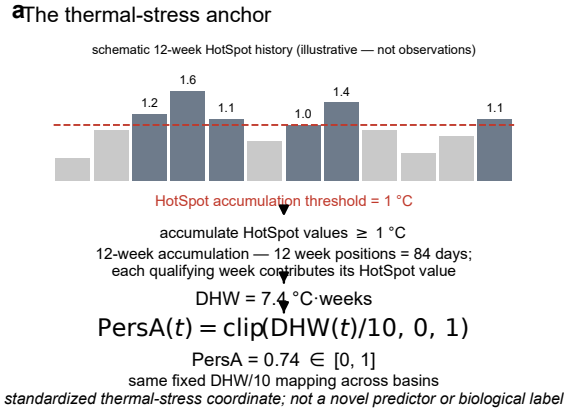
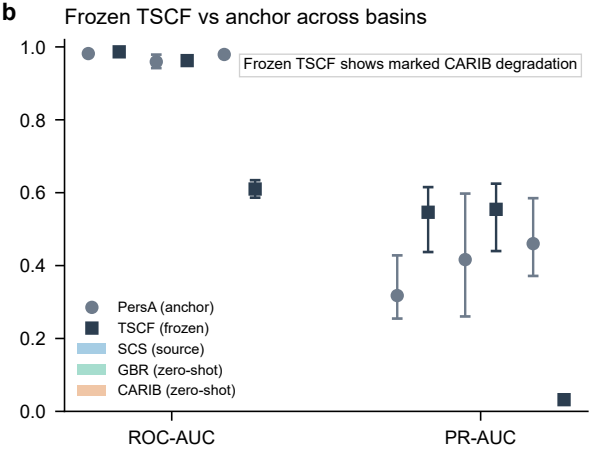
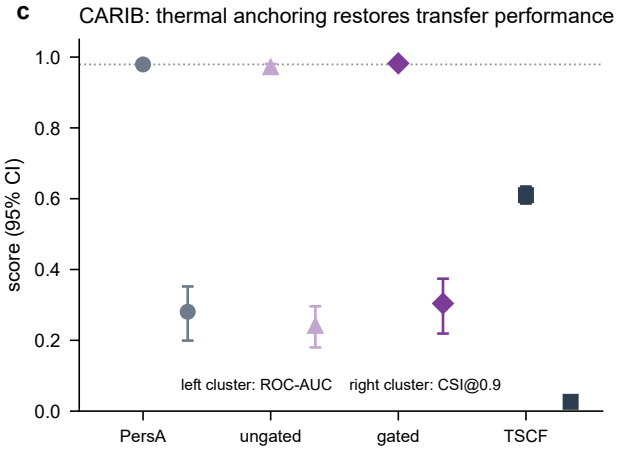
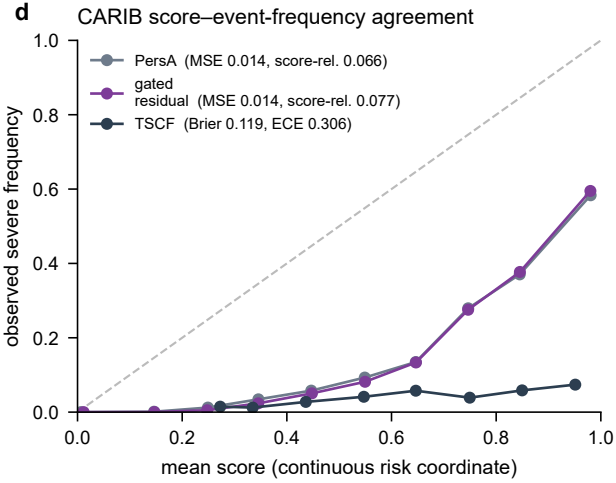

a
The thermal-stress anchor
schematic 12-week HotSpot history (illustrative — not observations)
1.2
1.6
1.1
1.0
1.4
1.1
HotSpot accumulation threshold = 1 °C
accumulate HotSpot values ≥ 1 °C
12-week accumulation — 12 week positions = 84 days;
each qualifying week contributes its HotSpot value
DHW = 7.4 °C·weeks
PersA(t) = clip(DHW(t)/10, 0, 1)
PersA = 0.74 ∈ [0, 1]
same fixed DHW/10 mapping across basins
standardized thermal-stress coordinate; not a novel predictor or biological label


b
Frozen TSCF vs anchor across basins
Frozen TSCF shows marked CARIB degradation
1.0
0.8
0.6
0.4
0.2
0.0
PersA (anchor)
TSCF (frozen)
SCS (source)
GBR (zero-shot)
CARIB (zero-shot)
ROC-AUC
PR-AUC


c
CARIB: thermal anchoring restores transfer performance
score (95% CI)
1.0
0.8
0.6
0.4
0.2
0.0
left cluster: ROC-AUC
right cluster: CSI@0.9
PersA
ungated
gated
TSCF


d
CARIB score–event-frequency agreement
PersA (MSE 0.014, score-rel. 0.066)
gated residual (MSE 0.014, score-rel. 0.077)
TSCF (Brier 0.119, ECE 0.306)
observed severe frequency
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
mean score (continuous risk coordinate)